

Introduction to Partitions

Discrete Mathematics, Durham University

Matthew Emerson

September 2026

Contents

1	Introduction	1
2	Introduction to Partitions	1
2.1	Basic Definitions	1
2.2	Bijjective Proofs	1
3	Visualising Partitions	1
4	Generating Functions	1
5	Computing $p(n)$	1
	Appendix	2

1 Introduction

Abstract

An introduction to partitions, with respect to discrete mathematics, as taught to me in my seminar sessions during my first year at Durham University. The following paper is as it was when it was originally submitted, with section one appended to the beginning.

The paper sets out the basic terminology of partitions, before diving into some of the applications and specific combinatorics problems that can be calculated using them.

Acknowledgements

In memory of the module professor, Professor Mikhail (Misha) Menshiokv, who passed away before the completion of this report.

2 Introduction to Partitions

2.1 Basic Definitions

Definition 2.1.0.1: A partition of $n \in \mathbb{N}$ is a way of writing n as the sum of numbers. These will be denoted in this document as $(p_1 + p_2 + \dots + p_i) \vdash n$.

Definition 2.1.0.2: A part is one of the numbers that make up a partition. It has an associated part size, which is the value of the number.

Definition 2.1.0.3: The partition function of n is the number of possible partitions of n . This will be denoted as $p(n) = i$.

Definition 2.1.0.4: The set of partitions is the set containing all possible partitions of n . This will be denoted using non-standard notation as $P(n) = \{p_1, p_2, \dots, p_i\}$ where p_x is a partition of n and $i = p(n)$.

Definition 2.1.0.5: Conditional partitions can also be created which limit which partitions are valid based on a given condition. This can be applied to both partition functions and sets of partitions as $p(n|condition)$ and $P(n|condition)$.

2.2 Bijective Proofs

Definition 2.2.0.1: A bijection is a mathematical function that is a one-to-one and onto mapping (Merriam-Webster, 2025).

3 Visualising Partitions

4 Generating Functions

5 Computing $p(n)$

Appendix

Merriam-Webster (2025). *BIJECTION Definition & Meaning - Merriam Webster*. [online] Merriam-webster.com. Available at: <https://www.merriam-webster.com/dictionary/bijection> [Accessed 9 Mar. 2026].